



QATIP Intermediate

Azure Lab03a

Service Principles

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Overview

This lab introduces secure, least-privilege access using **Azure Service Principals (SPs)**. You will:

- Create a Service Principal with limited permissions scoped to the subscription
- Use SP credentials via environment variables

- Run Terraform operations and observe RBAC restrictions
- Upgrade permissions and reattempt deployments

Before You Begin

1. Ensure you have Owner or User Access Administrator permissions at the **subscription** level.
2. Azure CLI and Terraform must be installed.
3. Navigate to the lab directory:

```
cd c:\azure-tf-int\labs\03a
```

Phase 0 – Create a Limited-Service Principal and Set Environment Variables

Purpose

What we are doing:

We are creating a new **non-human identity** (Service Principal) that will be used to run Terraform, representing a **limited automation identity** — like what you would use in CI/CD pipelines.

Why it matters:

Rather than using our full-access lab account, we want to simulate **realistic access boundaries** — mimicking how secure organizations prevent developers and automation tools from having unrestricted rights.

Environment variable usage:

Instead of hardcoding secrets, we inject them into the environment. This aligns with DevSecOps best practices and is supported natively by the Terraform AzureRM provider.

Steps

1. Navigate to **terraform.tfvars** and substitute <random unique suffix> with random letters and/or numbers of your choice to ensure name uniqueness.
2. Update line 2 of **main.tf**, replacing <your subscription id> with your lab subscription id.
3. Create the Service Principal scoped to your subscription (replace <sub_id> with your lab subscription and <3 digits> with 3 digits of your choice):

```
az ad sp create-for-rbac --name terraform-sp-<3 digits> --role Reader  
--scopes /subscriptions/<sub_id>
```

4. Capture the following values from the output:

```
appId → ARM_CLIENT_ID  
password → ARM_CLIENT_SECRET  
tenant → ARM_TENANT_ID  
subscriptionId → ARM_SUBSCRIPTION_ID
```

5. Set environment variables:

```
$env:ARM_CLIENT_ID = "<appId>"  
$env:ARM_CLIENT_SECRET = "<password>"  
$env:ARM_SUBSCRIPTION_ID = "<sub_id>"  
$env:ARM_TENANT_ID = "<tenant_id>"
```

6. Adopt the SP identity:

```
az logout  
  
az login --service-principal --username $env:ARM_CLIENT_ID --  
password=$env:ARM_CLIENT_SECRET --tenant $env:ARM_TENANT_ID
```

Then run:

```
az account show
```

You should see "type": "servicePrincipal" in the output.

Phase 1 – Run Terraform With Limited SP Identity

Purpose

What we are doing:

Attempting to deploy infrastructure using Terraform under this limited SP identity.

Why it matters:

This phase is expected to fail — and that is intentional. It highlights that simply authenticating to Azure is not enough; **role assignments define what you are allowed to do**. It also encourages developers to review error messages meaningfully and connect failures to permissions.

Steps

1. Run:

```
terraform init
terraform plan
terraform apply
```

2. Expect Terraform apply to **fail** with authorization errors.

Phase 2 – Assign Contributor Role to the Service Principal

Purpose

What we are doing:

Temporarily switching back to the lab account to assign Contributor rights to the SP.

Why it matters:

This demonstrates the **RBAC model in action** — where a principal's capabilities change dynamically based on role assignments. It mirrors how platform teams might approve temporary elevated access to unblock automation or development pipelines.

Steps

1. Re-adopt your lab identity:

```
az logout
```

```
az login
```

Enter your student credentials in the launched browser session

2. Inserting your SP id and Subscription id, run:

```
az role assignment create --assignee <appid> --role Contributor  
--scope /subscriptions/<sub_id>
```

3. Re-adopt the SP identity:

```
az logout
```

```
az login --service-principal --username $env:ARM_CLIENT_ID --  
password=$env:ARM_CLIENT_SECRET --tenant $env:ARM_TENANT_ID
```

4. Retry:

```
terraform plan
```

```
terraform apply
```

5. Terraform should now **succeed**.

6. Run **terraform destroy** to remove the resources.

7. Identify the App registration object id...

```
az ad sp show --id <appid> --query "{ objectId:id}" -o table
```

8. Attempt to delete the Service Principal...

```
az ad sp delete --id <object id>
```

9. Attempt to delete the App registration...

```
az ad app delete --id <appid>
```

10. The deletion attempts should fail because the Service Principal being used does not have sufficient permissions.

11. Re-adopt your lab identity:

az logout

az login

Enter your student credentials in the launched browser session

12. Delete the Service Principal: **az ad sp delete --id <object id>**

13. Delete the App registration: **az ad sp delete --id <appid>**

Congratulations, you have completed this lab

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